Beyond Recall: Operational Correctness for Agent Memory Systems

OpMem v0 Benchmark and Empirical Pins

Brainy Research

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Abstract

Retrieval benchmarks such as LoCoMo and LongMemEval measure whether an agent recalls the right fact, but not whether a memory *system* honors forget, correct, isolate, and supersede semantics in production. We present **OpMem v0**, a 13-task, pass/fail suite over neutral `remember`/`recall`/`revise`/`forget` scripts with substring-based assertions on ranked results. On the frozen 2026-08-15 pin (git SHA `1b5ab3e`), Brainy passes **13/13** via `GET /memories/search` while Mem0 **Platform** passes **10/13** (failures: `cor02`, `sup03`, `upd02`). This manuscript reports repository evidence, a reproducible protocol, threats to validity, and new embedded-harness re-measurements on the current merge-pin git revision where completed. We do *not* claim conversational-recall superiority over Mem0; OpMem operational leadership is orthogonal to LoCoMo J-scores.

1 Introduction

Production agents fail when suppressed memories leak after re-ingestion, corrections do not stick, tenants bleed across isolation boundaries, or stale facts outrank updates. Conversational QA benchmarks [3, 5] do not encode these lifecycle contracts; hallucination-oriented memory benchmarks such as HaluMem [2] focus on factual faithfulness rather than governed delete/correct semantics.

OpMem targets **operational correctness**: deterministic scripts, binary scoring, and explicit adapter mappings to each system's native API surface [1]. Section 8 lists threats; Section 6 reports primary, historical, and Phase 2 external-lane pins (Table 4).

2 Related work

Retrieval QA. LoCoMo [3] and LongMemEval [5] score end-to-end answer quality over long conversations; they are not substitutes for lifecycle tests.

Memory products. Mem0 documents accumulation-oriented designs and names staleness as an open problem [4]. We report Mem0 **Platform** numbers only on OpMem fixtures; we do not import Mem0 blog LoCoMo percentages as same-pin scores.

External memory lanes (OpMem v0). Mem0 **Platform** on OpMem v0 is **completed** at negligible API volume (13 tasks, 47 scripted steps; Section 6.8). Phase 2 counter-runs (after cost worksheet approval) are **completed** for Mem0 OSS, LangMem, Letta, and Zep; scores in Table 4: **9/13**, **8/13**, **5/13**, and **3/9*** respectively (Zep raw harness **3/13**; $0 paid). Paid or vendor-hosted lanes require a written estimate before approval; that gate was not a categorical denial of spend.

3 Neutral operation model

Actors are `(tenant, subject)` pairs. Tasks are operation scripts; `revise` and `forget` reference prior `remember` step ids. Adapters namespace tenants per task (`opmem-{nonce}-{task}`) for hermetic runs.

Table 1: Adapter mappings (normative; from `PUBLICATION_READINESS.md`).

Op	Brainy	Verbatim baseline	Mem0 Platform
remember	`POST /ingest`	append	`POST /v1/memories/`
recall	`GET /memories/search`	token overlap	`POST /v2/memories/search/`
revise	`POST .../correct`	in-place replace	`PUT /v1/memories/{id}/`
forget	`POST .../suppress`	delete	`DELETE /v1/memories/{id}/`

4 Task suite

OpMem v0 comprises 13 JSON fixtures under `fixtures/opmem/`: three suppression, three correction, three isolation, three staleness (including `upd03_state_supersession`), and one idempotency task. Assertions use case-insensitive substring checks on ranked `content` fields (`min_results`, `top_contains`, `must_exclude`, etc.).

5 Methodology

5.1 Scoring lane

Primary pins use Brainy `/memories/search`, not product `POST /recall` (Section 6.4 reports a `/recall` lane ablation). The harness (`evals/run_opmem.py`) exits non-zero only on infrastructure errors, not task failures—CI proves runner health, not 13/13, unless JSON is parsed.

5.2 Reproduction commands

```
go test ./internal/api/ -run TestOpMemBenchmarkAgainstHTTPServer -count=1
python3 evals/run_opmem.py --systems verbatim,brainy \
  --base-url http://127.0.0.1:8080 --json-out opmem-report.json
```

Paper experiments: `scripts/run-opmem-manuscript.sh` and `evals/run_opmem_paper_experiments.py`.

6 Results

6.1 Frozen 2026-08-15 pin (historical, SHA `1b5ab3e`)

[**completed**] Brainy **13/13**; verbatim **10/13**; Mem0 Platform **10/13** (artifacts `opmem-fresh-local-20260815.mo` `opmem-mem0-fresh-20260815.md`).

Mem0 failures on that pin: `cor02` (correction stickiness), `sup03` (durable forget leak), `upd02` (preference ranking).

Table 2: Category breakdown on 2026-08-15 Mem0 Platform counter-run.

Category	Mem0 Platform	Brainy (same cycle)
correction	2/3	3/3
isolation	3/3	3/3
suppression	2/3	3/3
staleness	2/3	3/3
idempotency	1/1	1/1
overall	**10/13**	**13/13**

6.2 Current-SHA re-measurement

[completed] At git SHA `20533f2ca83e` (UTC 2026-09-18T23:12:29Z), embedded API harness: Brainy (search lane) **13/13**, verbatim **10/13**. Mem0 Platform (same run): **10/13** (API surface: Platform via `Mem0OpAdapter`).

6.3 Combined machine-readable artifact

[completed] Committed `opmem-primary.json` lists per-task outcomes for verbatim, brainy, mem0 with manifest fields (SHA, date, fixture count).

6.4 Recall vs search lane

Table 3: Brainy recall lane and revise mapping (SHA 20533f2ca83e).

Configuration	Overall
brainy	13/13
brainy-recall	9/13
brainy-supersede	11/13

Search (`/memories/search`) is the normative pin lane. `brainy-recall` uses `POST /recall` (context mode). Correction tasks on the recall lane may return HTTP 409 during `/correct` (infrastructure errors in artifact JSON).

6.5 Correct vs supersede mapping

[completed] Mapping `revise` → `POST .../supersede` yields **11/13** vs **13/13** with `/correct` on the search lane (same SHA). Failures are task-level assertion misses, not missing runs.

6.6 Mechanism breakdown

Category pass rates separate correction/suppression/staleness contracts; verbatim approximates rank-only recall without governed lifecycle ops.

6.7 Statistical stability

[completed] Five consecutive Brainy search-lane repeats on one embedded server session. **Valid runs only** (infra errors = 0): 2/5; scores: 13/13.

Category	Brainy	Verbatim (rank-only)
correction	3/3	2/3
idempotency	1/1	0/1
isolation	3/3	3/3
staleness	3/3	3/3
suppression	3/3	2/3

- Repeat 1: 13/13 (infra errors=0)
- Repeat 2: 9/10 (infra errors=3)
- Repeat 3: 9/10 (infra errors=3)
- Repeat 4: 13/13 (infra errors=0)
- Repeat 5: 9/10 (infra errors=3)

Repeats with infra errors > 0 (3 of 5) are **invalid/incomplete**: the harness omits errored tasks from the denominator, so aggregates such as 9/10 are *not* comparable to a clean 13/13 pin. Use isolated `TestOpMemBenchmarkAgainstHTTPServer` for merge-gate stability.

6.8 External systems

[**completed**] Mem0 **Platform**: 10/13 on SHA `20533f2` (`opmem-pin-20260918.json`); same headline as 2026-08-15 pin. Completed under negligible Platform API use (full suite ≈47 scripted steps).

Lane	OpMem score	Run
Brainy (primary pin)	13/13	`opmem-pin-20260918`
Mem0 Platform	10/13	same pin
Mem0 OSS	9/13	`opmem-phase2-full-mem0`
LangMem	8/13	`opmem-phase2-full1`
Letta	5/13	`opmem-phase2-letta-full`
Zep	3/9*	`opmem-phase2-zep-full`

Table 4: Phase 2 external lanes: each lane ran the full 13-task OpMem script; Zep is scored on **9 supported tasks***. *Four forget fixtures (`iso03_forget_isolated`, `sup01--sup03`) are *not supported* on the Zep lane; raw harness tally **3/13** unchanged on `live-full-opmem-phase2-zep-full.json`. Forget is not supported (`zep-cloud` 3.28 thread/graph API has no delete path the adapter can use).

[**completed**] Mem0 **OSS** (local Qdrant path, `mem0ai` 2.1.0): **9/13** on run `opmem-phase2-full-mem0` (0 infra errors). Artifact: `live-full-opmem-phase2-full-mem0.json`. [**completed**] LangMem (Postgres 16 + `pgvector` 0.6): **8/13** on run `opmem-phase2-full1` (0 infra errors). Artifact: `live-full-opmem-phase2-full1.json`. [**completed**] Letta (`letta==0.11.7`): **5/13** on run `opmem-phase2-letta-full` (0 infra errors). Artifact: `live-full-opmem-phase2-letta-full.json`. [**completed**] Zep ($0 paid, `zep-cloud` 3.28.0): **3/9*** on supported tasks from run `opmem-phase2-zep-full` (0 infra errors). *Not supported* (not scored as failures): `iso03_forget_isolated`, `sup01_basic_forget`, `sup02_targeted_forget`,

sup03_durable_forget. Raw harness: **3/13**.
Artifact: live-full-opmem-phase2-zep-full.json; interpretation: zep-score-interpretation-opmem-phase2-zep-full.json. [**completed**] Payload budget worksheet: worst-case OpenAI **$0.15** vs $3 ceiling (cost-worksheet-payload-budget.json).

6.9 Cost-estimation plan (before any new paid run)

OpMem v0 has 13 tasks and 47 scripted steps (fixtures/opmem/). Before any new paid lane run, prepare billing and approval:

- **Phase 1–2** — evals/opmem_external/: deterministic namespaces, per-task reset, payload budget (cost-worksheet-payload-budget.json), live harness run_opmem_external_live.py. Setup: external-lanes-setup.md.
- **Mem0 OSS** — Local Qdrant + OPENAI_API_KEY; ceiling $1; dry-run projected OpenAI spend in worksheet.
- **Zep** — Zep Cloud free tier only ($0 paid); cannot pin OpenAI models (product constraint recorded).
- **Letta** — Self-hosted App Server; ceiling $1; no legacy Docker API or Letta Cloud paid plan.
- **LangMem** — Postgres/pgvector; ceiling $1; no InMemoryStore on measured runs.

No dollar totals are asserted here until rate cards and deployment mode are pinned; approval requires the completed worksheet, not a categorical spend ban.

7 Discussion

OpMem failures on Mem0 align with accumulation-oriented update semantics: corrections and durable forgets are not first-class lifecycle events. Verbatim baseline passes 10/13, showing the suite discriminates rank-only stores from governed suppression/correction. Policy disagreements remain (sup03 durable forget is a normative contract).

8 Limitations

1. Small $n = 13$; each task is $\sim$7.7% of the aggregate score.
2. Substring scoring does not guarantee semantic safety.
3. Search vs /recall product path may diverge (lane ablation quantifies on Brainy).
4. English, domain-neutral fixtures; no concurrency or TTL tests.
5. Mem0 Platform API drift; Phase 2 OSS/LangMem/Letta/Zep pins are single-pass, small n. Zep forget fixtures are *not supported* on the lane (Table 4 footnote); publication line **3/9*** vs raw harness **3/13**.
6. Fixture authorship bias; no independent external audit documented.

9 Artifact appendix

Committed run manifests and JSON under `docs/benchmarks/artifacts/opmem-manuscript-*`: `run-manifest.json`, `opmem-primary.json`, `opmem-lane-ablation.json`, `opmem-stability.json`, `opmem-mechanism.json`, and `commands.log`.

Build PDF: `make -C docs/research/opmem/paper pdf`.

10 Future work

v1 task expansion (∼30 tasks), Zep forget/delete mapping if a supported API path appears, supersede-native revise mapping as default where lineage is required, and fail-closed CI assertion of 13/13 on merge pins.

References

[1] Brainy. OpMem v0 specification and harness. `https://github.com/tryvinci/brainy` (`docs/research/opmem-spec.md`), 2026.

[2] Ding Chen, Simin Niu, Kehang Li, Peng Liu, Xiangping Zheng, Bo Tang, Xinchi Li, Feiyu Xiong, and Zhiyu Li. HaluMem: Evaluating hallucinations in memory systems of agents. *arXiv preprint arXiv:2511.03506*, 2025.

[3] Adyasha Maharana et al. LoCoMo: Evaluating very long-term conversational memory of llm agents. *arXiv preprint arXiv:2402.17710*, 2024.

[4] Mem0. State of AI agent memory (2026). `https://mem0.ai/blog/state-of-ai-agent-memory-2026`, 2026. Vendor context on staleness and cross-session evolution; not an OpMem same-pin score.

[5] Howard Yen Wu et al. LongMemEval: Benchmarking long-term memory in llm chat assistants. *arXiv preprint arXiv:2410.10819*, 2024.